

NTUST course: Computer Vision and Applications (CI5336701, 2026 Spring)

Final Project : Reconstruct 3D points based on triangular method

Date Due : 2026. Jun. 12th, PM11:59 (2.5 weeks)

Description :

1. Write a program for reconstructing 3D points a sequence of images, then, output all 3D color points as one XYZ file. (choose your tools, ex. C++/C, python, openCV, or Matlab).
2. This is one scenario to simulate a 3D laser scanner consisting of a slit laser and one hexagonal turntable. The hexagonal turntable has a diameter of 50 mm and a thickness of 15 mm. It rotates about the z axis, and the laser beam passes through its axis of rotation. The camera is always at the same position, and its intrinsic parameter K is known as:

$$\begin{bmatrix} 720 & 0 & 360 \end{bmatrix}$$
$$\begin{bmatrix} 0 & 720 & 640 \end{bmatrix}$$
$$\begin{bmatrix} 0 & 0 & 1 \end{bmatrix}$$

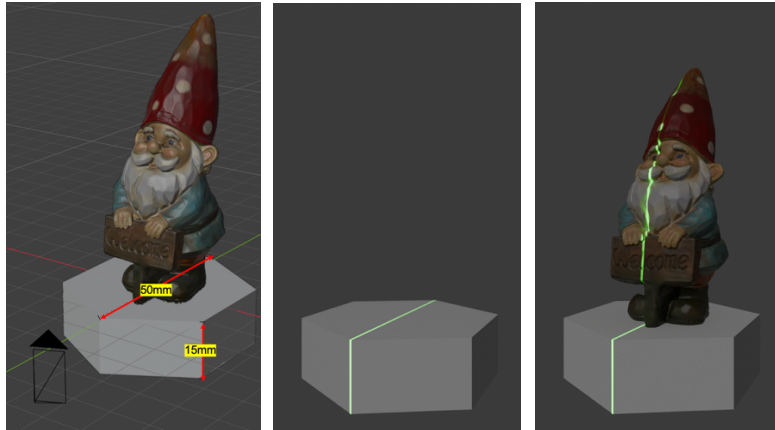
3. In each frame, as the hexagonal turntable rotates, the 3D model on the hexagonal turntable undergoes a rotational motion, with each frame captured at two-degree intervals. During operation, a green slit laser continuously casts a line across the surface of the 3D model. Please detect the green slit laser in each image and reconstruct a 3D point cloud from all the laser pixels across the images. (Both types of images are given. You can detect green line by either your own method or image subtraction).
4. Please write a short report (up to 4 pages in A4), and try to visualize your result and provide the error map (ex. using “cloudcompare” software) to proof the performance of your estimation.
5. Deliverable: There are three items you should provide:
 - 1) Source code (and execution file, if any) in python, C++/C, Matlab, with simple comments,
 - 2) An output 3D file in YourID.xyz (Color 3D point cloud),
 - 3) report document (4 pages at most).

Please zip all your files, then, upload to moodle2 (<http://moodle2.ntust.edu.tw>) by date due.

Note: this assignment will be **15%** of final grade.

Hint:

1. Please refer to course slides.
2. The configuration looks like the following figure:



3. Here is the similar concept on Cellphone app: <https://youtu.be/JL6SF2gQ35o>

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